



QUICKSTART

Photogrammetry Model Creation
Final Presentation

Thursday
07/02/2026



Team Overview

□ Sebastian Noel



Role: RealityScan Model Creator

□ Gavin Heaver



Role: Photographer and Initial RealityScan Tester

□ Stevin George

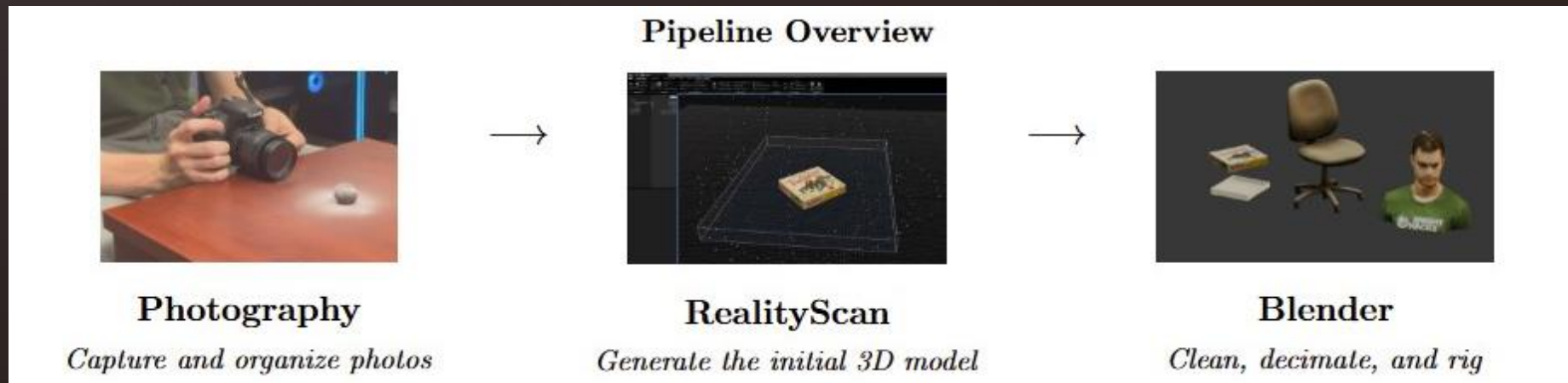


Role: Unity/Blender Rigger



Project Description

- This project tested the photogrammetry software RealityScan to create riggable 3D models that can be used within Unity developed VR games. The models needed to vary in size and shape, being small (Chestnut), medium (Game Box), large (Cabinet/Chair), extra large (Room), and person.
- The project was to illustrate a detailed pipeline for taking photos of a variety of objects, convert the photos into a 3D model using RealityScan, rig the model for basic functionality, and import the models into Unity.



Project Goals/Limitations

□ Goals:

- Create a detailed pipeline that displays the inputs and outputs at each major point, going from photography into RealityScan into Blender into Unity
- Within the pipeline, the goal was to keep things convenient without sacrificing any quality, while also keeping the model renderings computationally inexpensive (i.e., limit polygon count)

□ Limitations:

- Avoid AI and cloud usage, as both act as security risks
- Do not use RealityScan App due to image processing coming from Epic Games' cloud servers
- Use only photographs for photography model creation (i.e., no LiDAR)



Technology Stack

□ Software/languages:



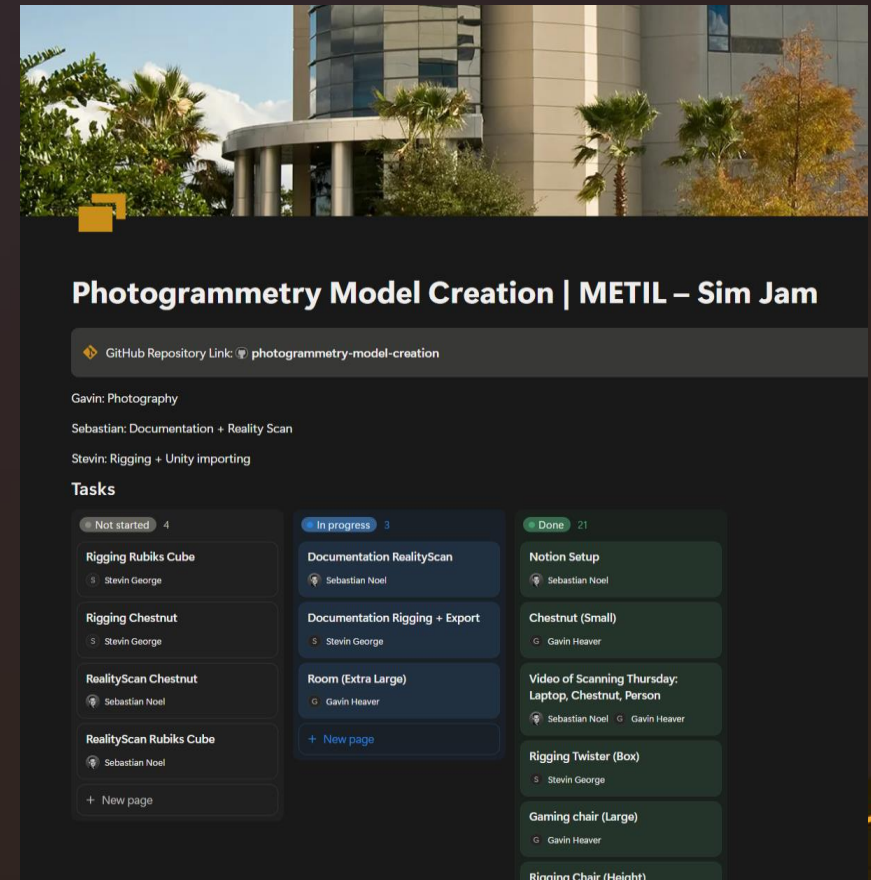
□ Hardware:

- Computer with Nvidia GPU (until recently)
- DSLR/Mirrorless Camera Setup
 - Camera, 20-50mm lens, SD card, tripod, table to support items, and lighting

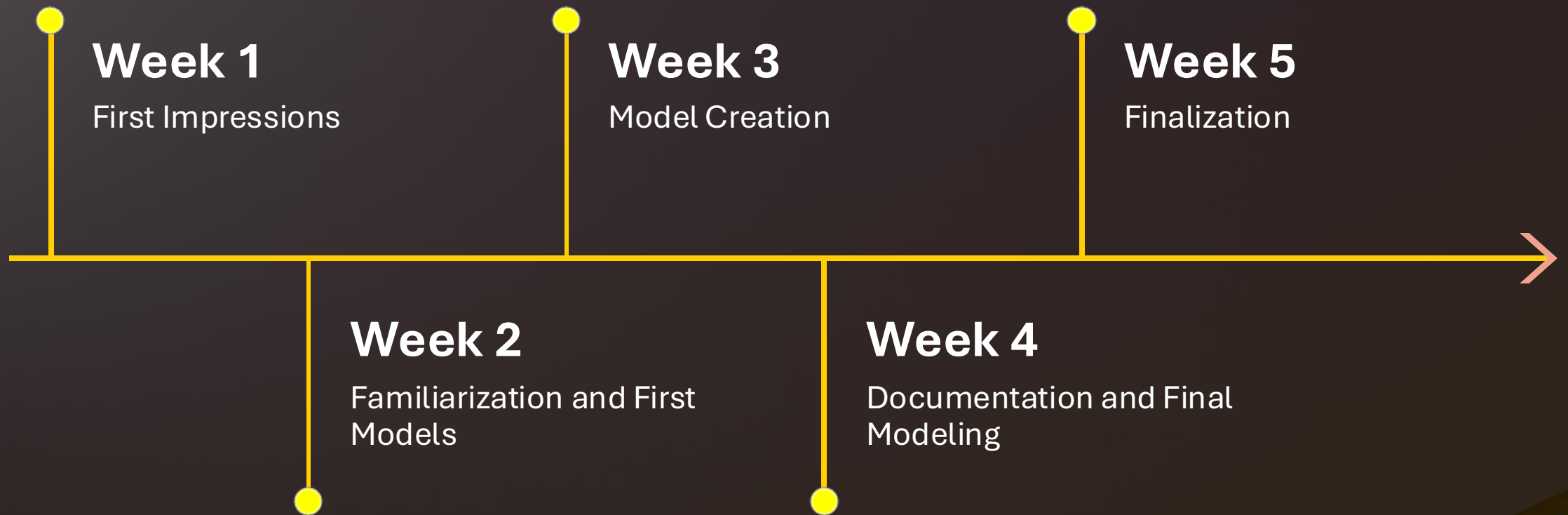


Approach

- Dividing up tasks into evenly split roles (Notion)
- Initial research to understand basic requirements
 - Youtube videos for photography and RealityScan:
 - <https://www.youtube.com/watch?v=3mLBL7T-zgs>
 - <https://www.youtube.com/watch?v=4qFl4k37dDc&t=689s>
- Weekly meetings
 - Discuss what we have done, what we still need to do, and set goals to reach for the upcoming week
- Experiment with tools and software
 - Documenting anything that worked or didn't work, along with why
- Clearly setting the inputs and outputs for each portion of the pipeline



Project Plan



Task Breakdown – Photography

- Task: Take photos of objects following all requirements with the given methods that can be used and combined in RealityScan
- 150 – 200+ photos
- Photography Requirements:
 - Flat lighting
 - Consistency
 - Object fills 70-80% Frame
 - 60-80% Overlap (any point should appear at least 5 times at different angles)



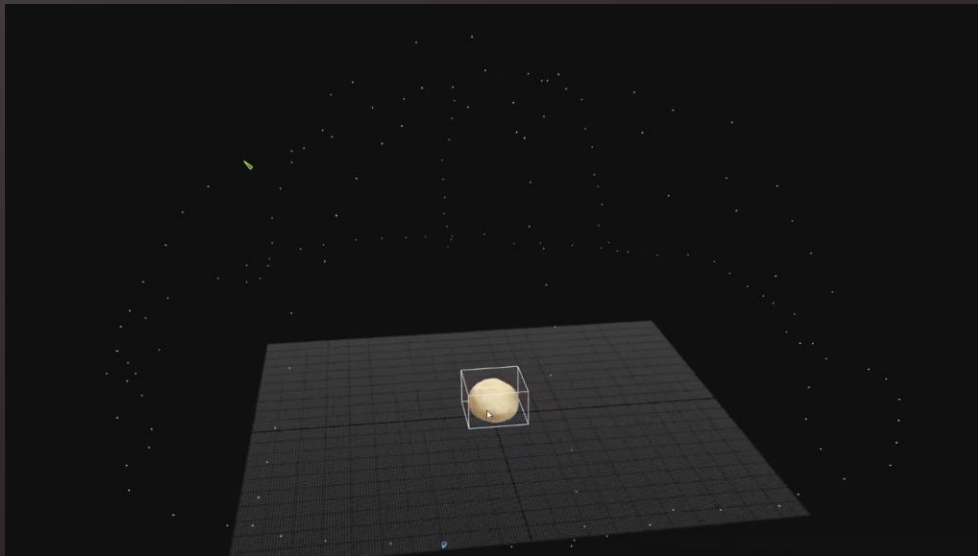
Task Breakdown – Photography

- Camera Setup
 - Camera Mode: Manual
 - Aperture: f/8 – f/11
 - ISO: 100
 - Shutter Speed 1/125s
 - White Balance: manual
- Lowlight compromises (interact in this order)
 - Lower shutter speed, not below 1/40s – 1/60s
 - Raise the ISO
 - Widen aperture to a max of f/6 to avoid warping



Task Breakdown – Photography

- Method 1 (Crossing over):
Great for small/medium objects



- Method 2 (Multi Layered):
Perfect for large, extra large, and humans



Note: if every rule is followed, you could combine methods to capture the most details possible



Task Breakdown - Photography

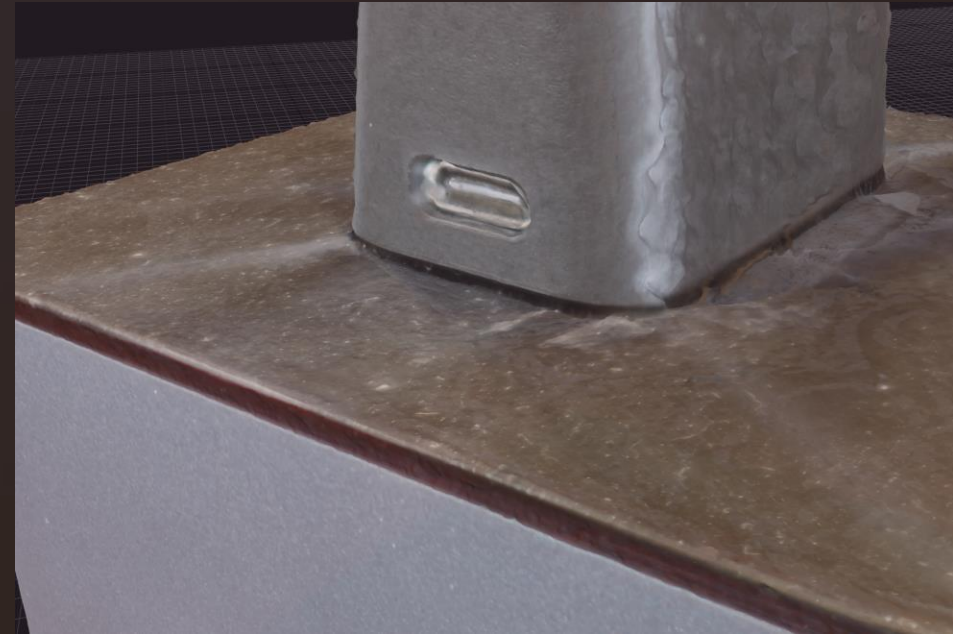
□ Glare

- Can be reduced with scanning spray or dry shampoo

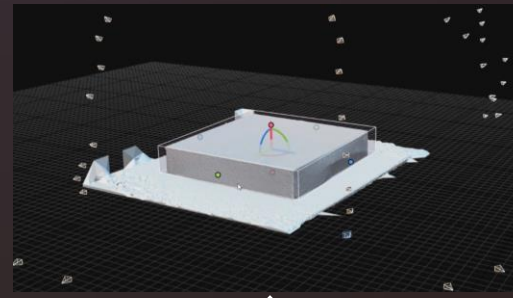
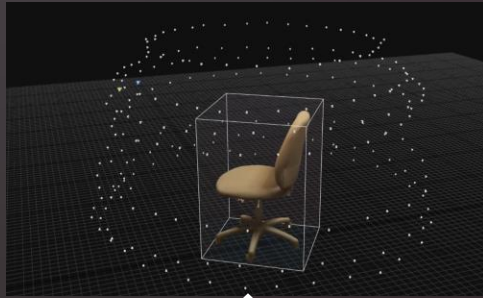


□ Holes

- Just doesn't come across well in photos



Task Breakdown - RealityScan



Importing
Images

Aligning
Images

Reconstruction
Region

Mesh in
Normal Detail

Cleanup

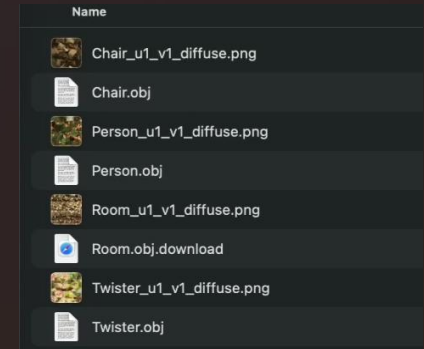
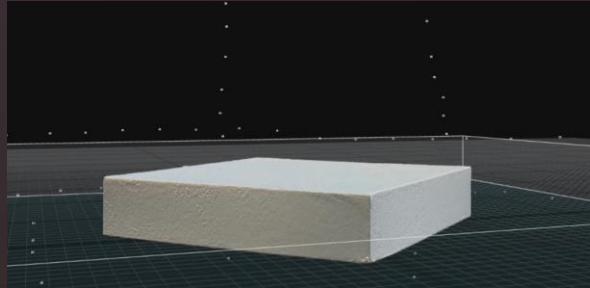
Exporting
Masks

Then simply drag and drop
into RealityScan

Tighten the region bounds to contain
only the object you want



Task Breakdown - RealityScan



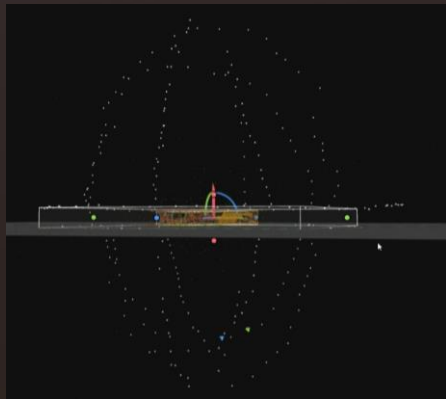
Repeat for
other side

Aligning both
sides

Region, Mesh,
Cleanup

Texturize

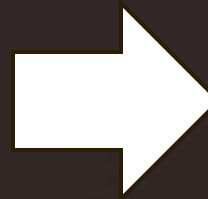
Export



Task Breakdown – Blender/Unity

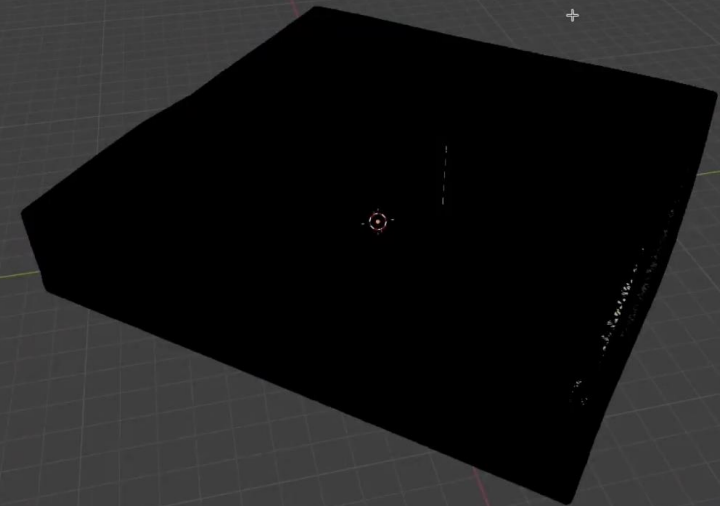
□ Step 1: Cleaning up your Model

Weight Paint 1 / 5,318,741
Texture Paint 22 / 15,956,222
Faces 10,637,477 / 10,637,477
Triangles 10,637,477 / 10,637,477



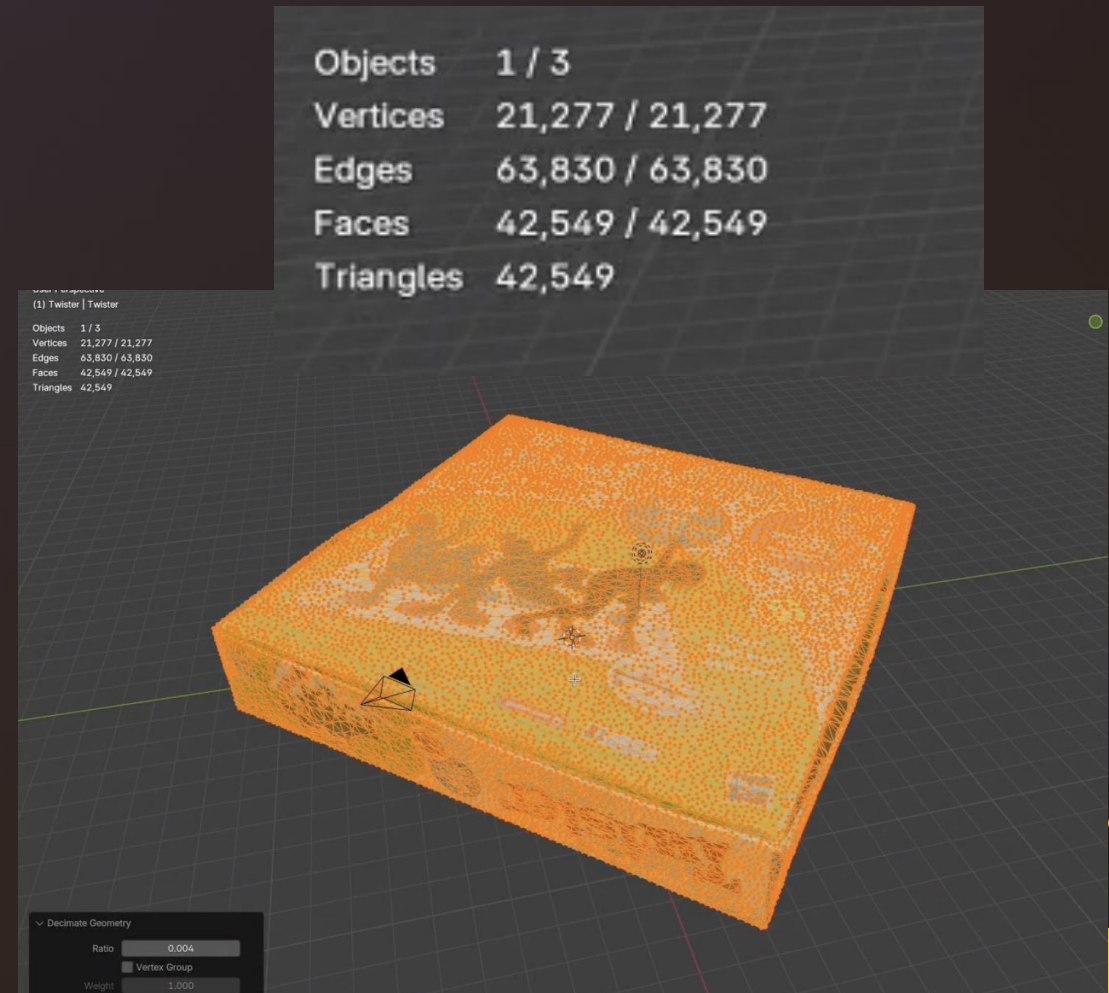
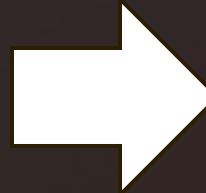
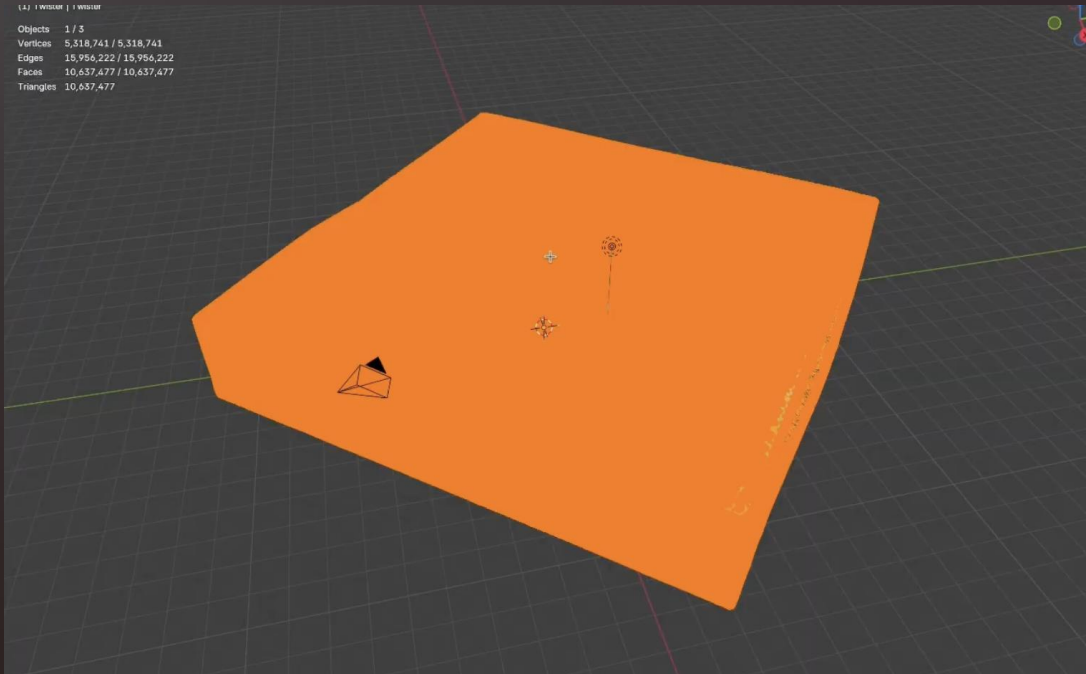
Objects 1 / 3
Vertices 0 / 5,318,741
Edges 0 / 15,956,222
Faces 0 / 10,637,477
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Vertices 0 / 5,318,741
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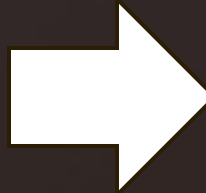
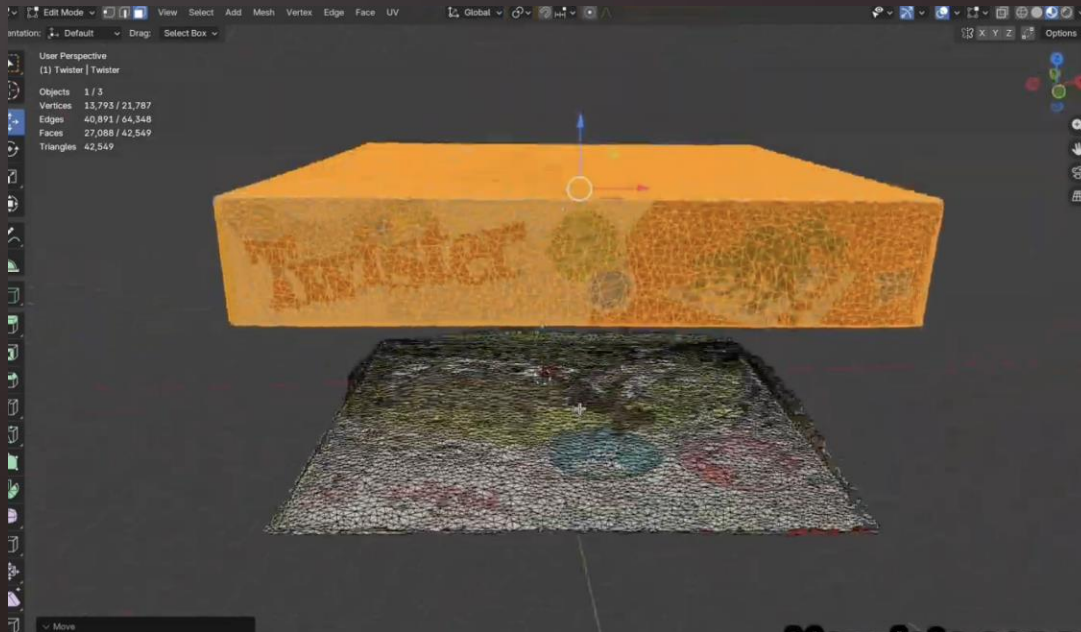
Task Breakdown – Blender/Unity

□ Step 1: Cleaning up your Model



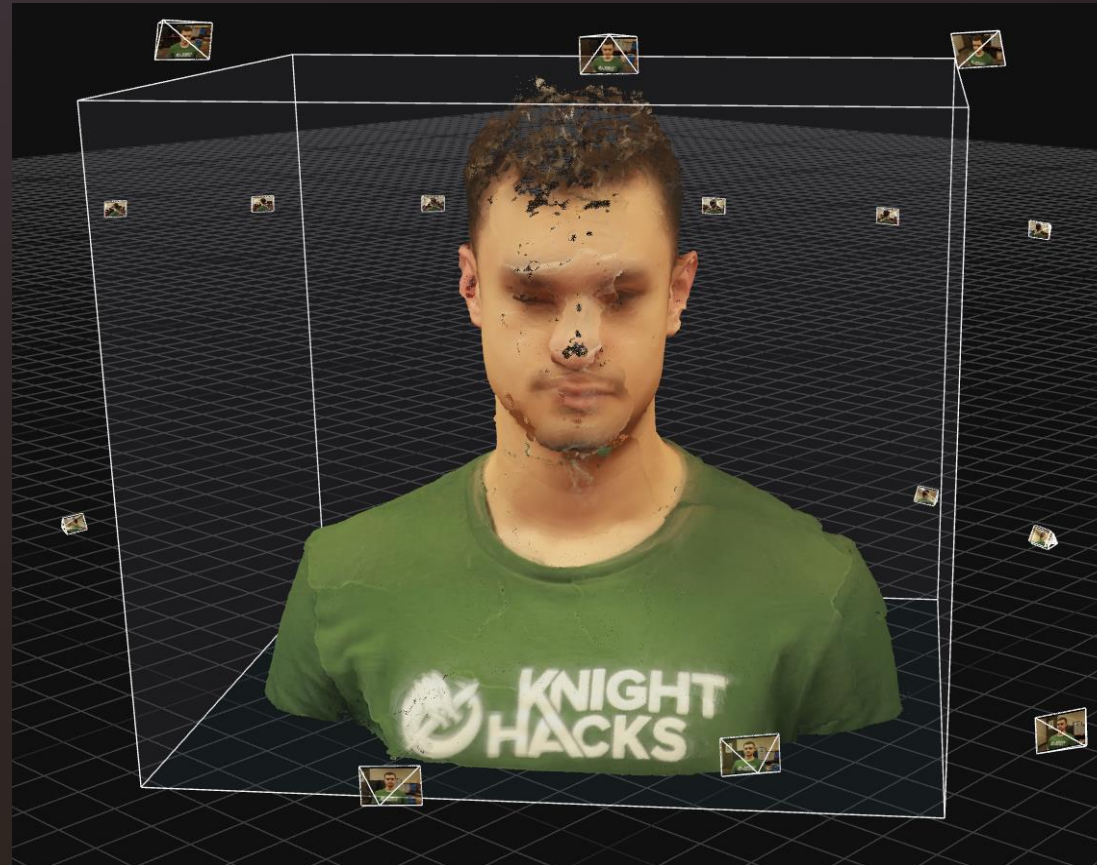
Task Breakdown – Blender/Unity

□ Step 2: Simple Rigging



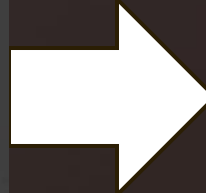
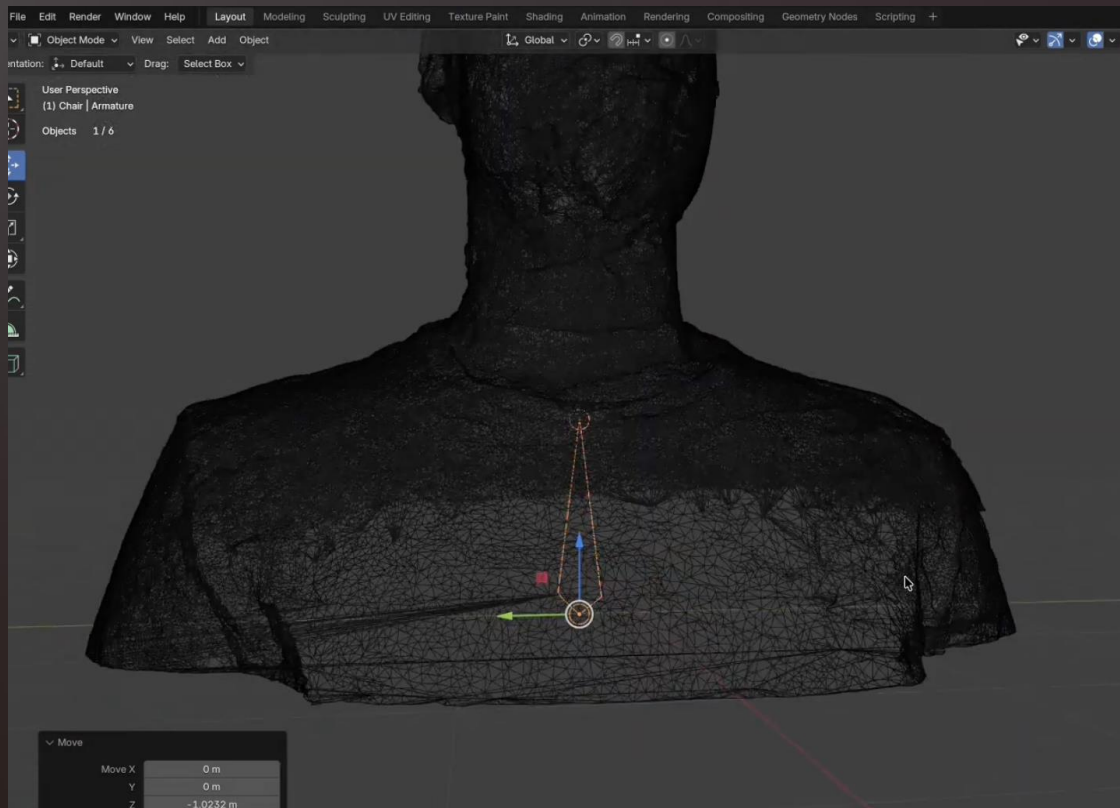
Task Breakdown – Blender/Unity

□ Step 2: Armature Rigging



Task Breakdown – Blender/Unity

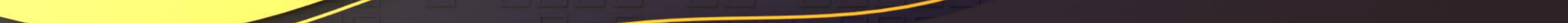
□ Step 2: Armature Rigging



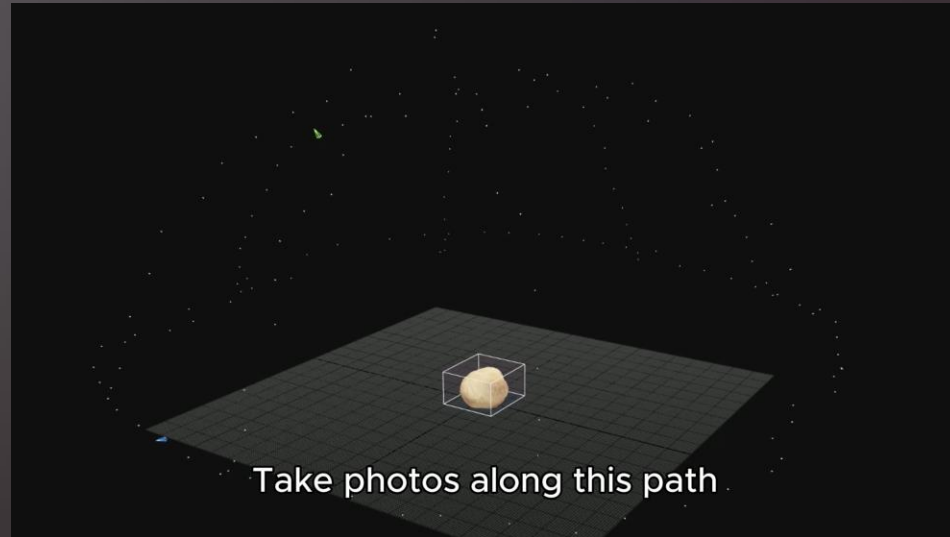
Task Breakdown – Blender/Unity

Step 3: Importing to Unity





Project Demonstration

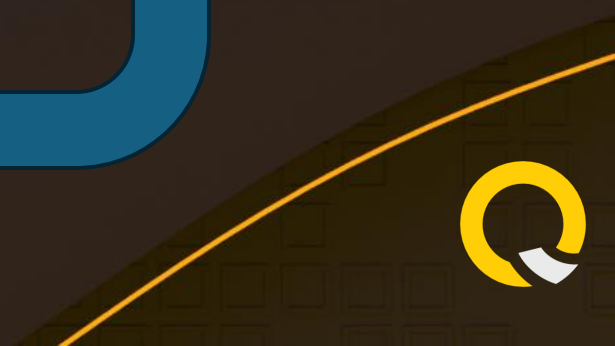
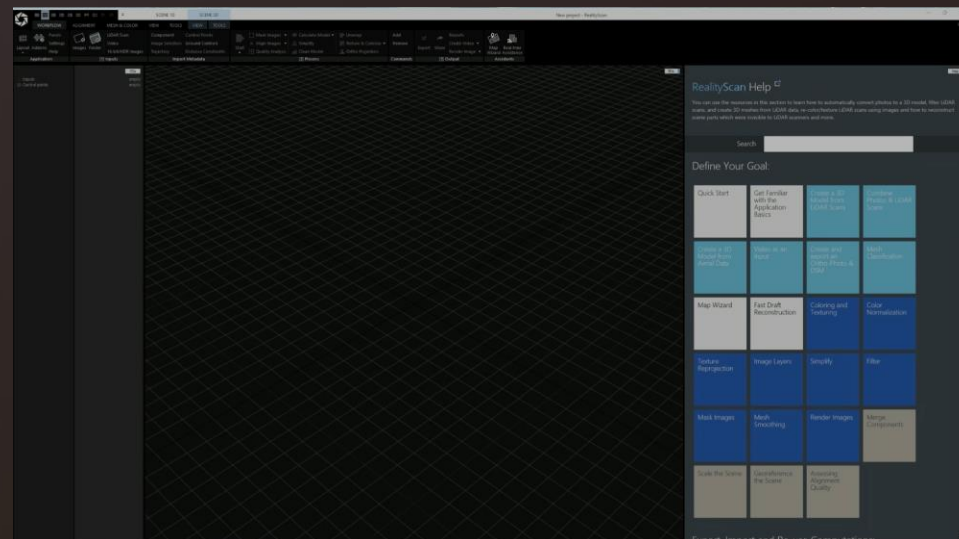


RealityScan

1. Import .obj model and .png textures from RealityScan

Blender

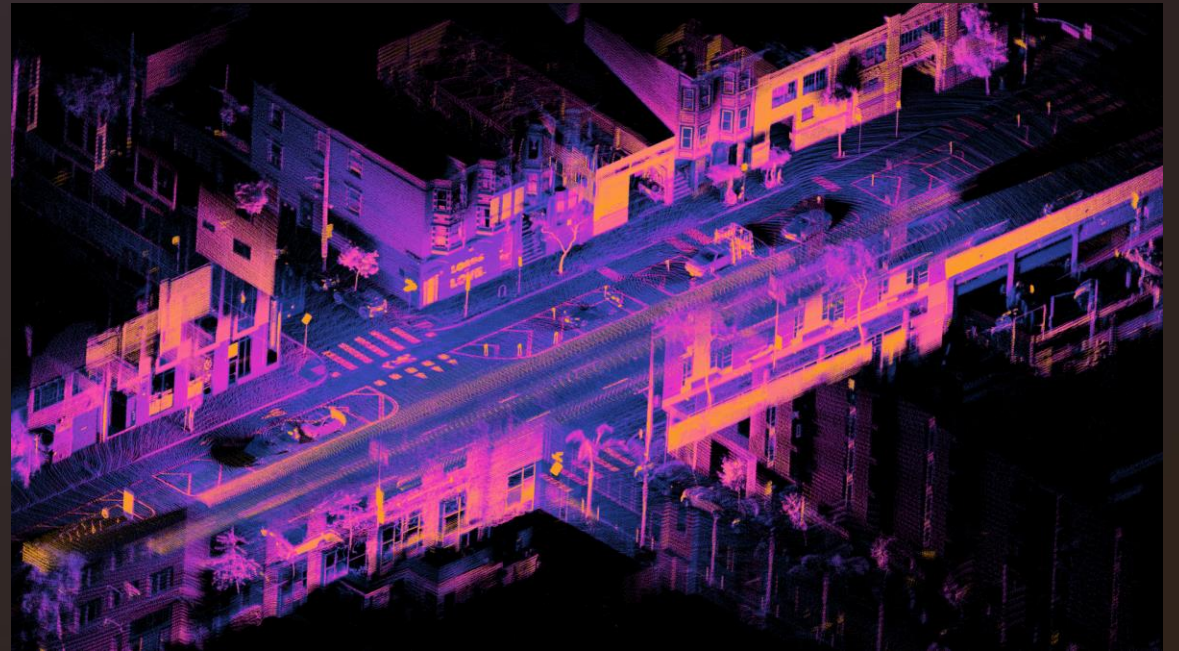
Photography





Project Next Steps

1. LiDAR with RealityScan
2. Photography Setups
3. Trying different photogrammetry software (RealityScan Mobile, Meshroom, COLMAP)



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□ Lessons Learned:

- Project Management
- Communication is key

□ Road Blocks/Hurdles:

- Identifying best photos for RealityScan
- Merging top and bottom models in RealityScan
- Managing poly-count in Blender/Unity

□ What we would do differently:

- Finding photography techniques to use turntable
- Using control points within RealityScan more to stitch bottom and top scans without needing more photos

Lessons Learned

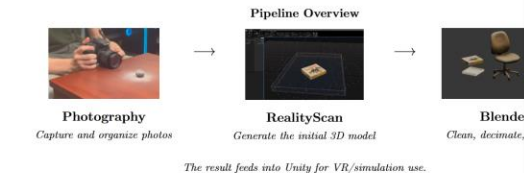
Photogrammetry Model Creation

UCF Institute for Simulation & Training | Quickstart SimJam

Gavin Heaver • Sebastian Noel • Stevin George

Abstract

This project documents a repeatable pipeline for turning ordinary, flat (2D) photographs into textured, colored 3D models that are ready for use in VR and simulation applications. Using RealityScan as the core photogrammetry tool, the process moves from photography, to model generation, to rigging in Blender, and finally to import into Unity. A deliberate constraint of the project was to avoid AI and cloud-based tools, which act as security risks, while still keeping the workflow convenient and the resulting models lightweight enough to remain computationally inexpensive. We found that high-quality results are achievable with accessible equipment (a single DSLR/mirrorless camera, controlled lighting, and disciplined photography), provided a consistent set of rules is followed. The workflow is organized by object scale (small/medium objects, large objects, rooms, and people) so that it can serve as a practical, step-by-step guide from the first Unity-ready model.



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Capturing the room. Make sure windows aren't visible (shut curtains or cover them) and that nothing is moving, with all loose objects secured.

1. Stand in the middle of the room (or the area you're capturing). Take a photo, turn slightly left by about 5-10°, and take another. Repeat for a full 360°.
2. Angle the camera up to capture the upper walls and roof, again a photo every 5-10° for a full 360°.
3. Angle the camera down to capture the lower walls and floor, again a photo every 5-10° for a full 360°.

At this point a room model can be rendered to act as a starting point for mimicking rooms.



Figure 9. Three-pass room capture from a central standing position (left to right: angled down, level, and angled up).

Adding detail (optional). For more detail, take photos above/in front of the furniture, circling the room with 90-80% overlap per photo so RealityScan knows where each photo is angled relative to the others. If you want something specific captured (e.g., the inside of an open bag), photograph everything in and around it, with transitions between the "major" angles. Don't be afraid to capture from slightly different angles to show how an object looks on all sides.





Questions?

